TryHackMe Writeup: Introductory Networking

- **Platform:** TryHackMe
- **Room:** https://tryhackme.com/room/introtonetworking
- **Topics:** OSI and TCP/IP models, encapsulation, TCP handshake, ping, traceroute, WHOIS, DNS and dig

1. Objective

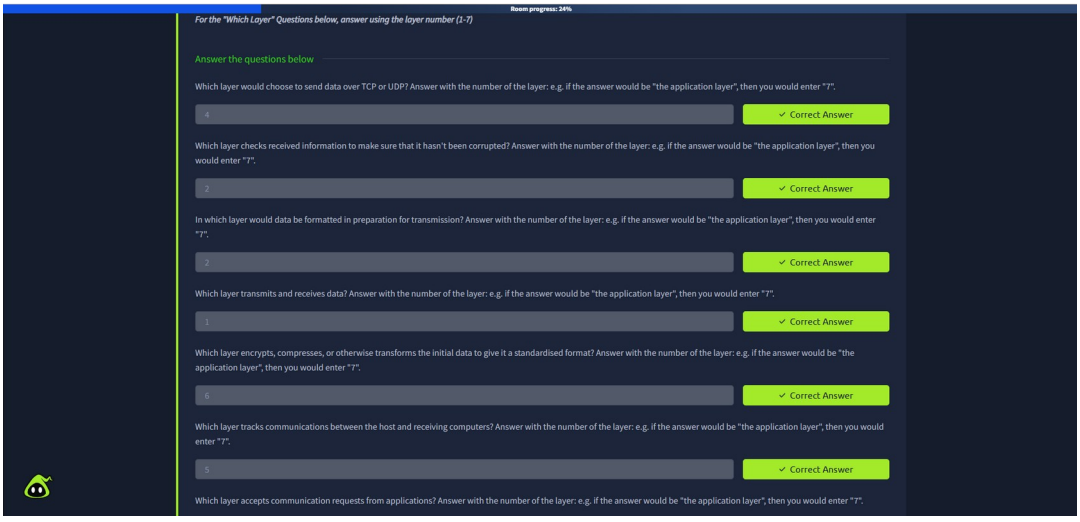
This room introduces networking models and basic tools used to understand connectivity and investigate network information. It covers the OSI model, TCP/IP, data encapsulation, TCP connection setup, ping, traceroute, WHOIS and DNS lookups.

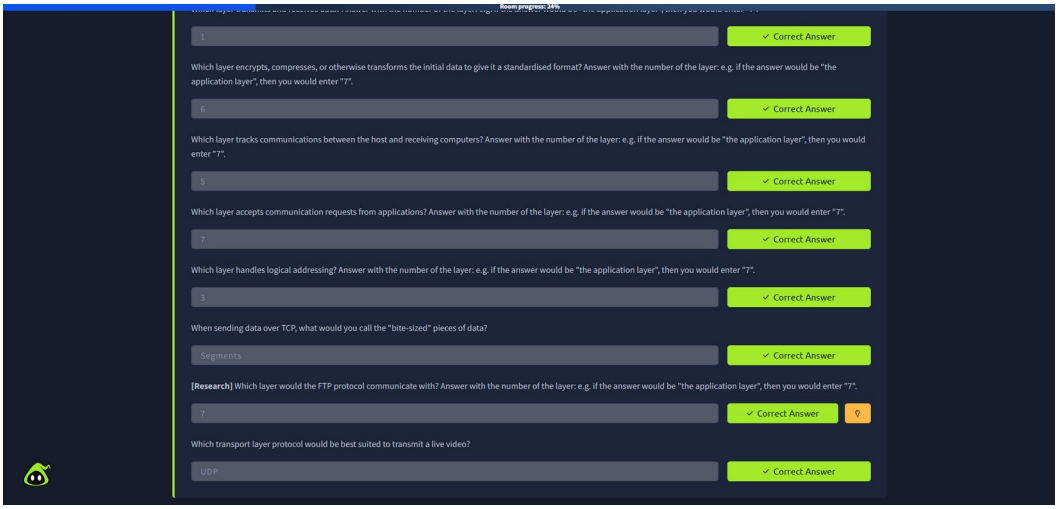
2. Task and Methods

Task 2 - The OSI Model

The OSI model has seven layers, from Layer 1 (Physical) to Layer 7 (Application).

OSI layer	Number	Main purpose
Application	7	Provides network services to applications.
Presentation	6	Formats, compresses, and may encrypt data.
Session	5	Establishes and manages communication sessions.
Transport	4	Uses TCP/UDP and divides data into transport units.
Network	3	Logical addressing and routing using IP.
Data Link	2	Local frame delivery and error checking.
Physical	1	Transmits raw bits as electrical, optical, or radio signals.





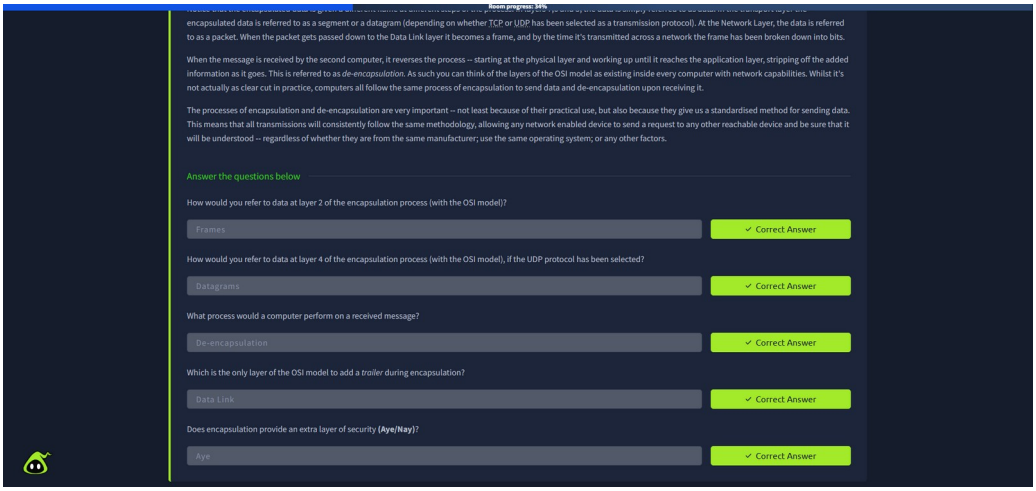
Answers

Question	Answer
Which layer chooses TCP or UDP?	4
Which layer checks received information for corruption?	2
Which layer formats data for transmission?	2
Which layer transmits and receives signals at the physical medium?	1
Which layer encrypts, compresses, or transforms data into a standard format?	6
Which layer tracks communications between hosts?	5
Which layer accepts communication requests from applications?	7
Which layer handles logical addressing?	3
What are TCP's bite-sized pieces called?	Segments
Which OSI layer does FTP operate at?	7
Which transport protocol is often preferred for live video?	UDP

Task 3 - Encapsulation

Encapsulation adds protocol-specific information as data passes down networking layers before transmission. At the receiving device, **de-encapsulation** removes and processes those layers in reverse order.

Layer / stage	Name of data
Application, Presentation, Session	Data
Transport with TCP	Segment
Transport with UDP	Datagram
Network	Packet
Data Link	Frame
Physical	Bits



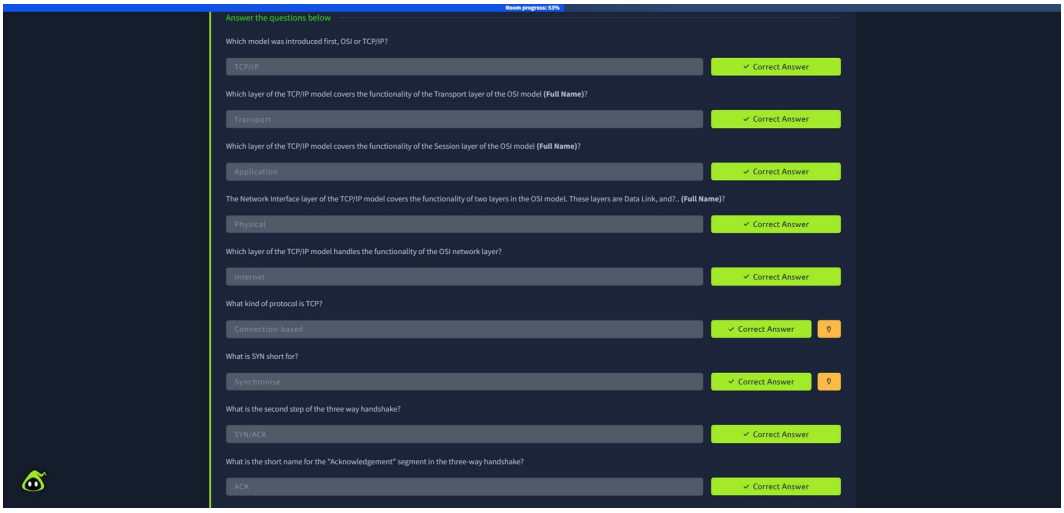
Answers

Question	Answer
What is data called at OSI Layer 2?	Frames
What is data called at Layer 4 when UDP is selected?	Datagrams
What process is performed on a received message?	De-encapsulation
Which layer adds a trailer?	Data Link
Does the room's lesson describe encapsulation as adding an extra layer of security?	Aye (room's expected wording)

Technical note: Encapsulation structures delivery and some protocols include error-detection fields, but encapsulation by itself is not encryption and does not guarantee security against interception or tampering.

Task 4 - The TCP/IP Model

The common four-layer TCP/IP model consists of Application, Transport, Internet and Network Interface. It groups some functions differently from the seven-layer OSI model.

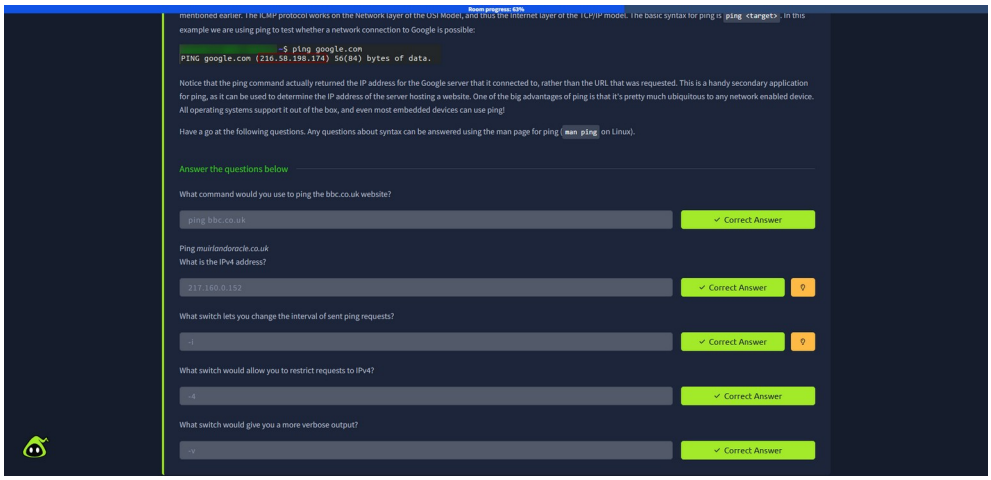


Question	Answer
Which model was introduced first?	TCP/IP
TCP/IP layer corresponding to the OSI Transport layer	Transport
TCP/IP layer covering the OSI Session layer's functionality	Application
Other OSI layer covered by Network Interface, along with Data Link	Physical
TCP/IP layer corresponding to the OSI Network layer	Internet
What kind of protocol is TCP?	Connection-based
What does SYN stand for?	Synchronise
What is the second step of the TCP three-way handshake?	SYN/ACK
Short name for acknowledgement	ACK

The TCP three-way handshake is **SYN → SYN/ACK → ACK**. TCP establishes a connection and supports reliable, ordered delivery.

Task 5 - Networking Tool: Ping

ping sends probe messages to check whether a destination responds and to estimate round-trip time. It commonly uses ICMP.



Question	Answer
Command to ping BBC's website	ping bbc.co.uk
IPv4 address of muirlandoracle.co.uk in room walkthroughs	217.160.0.152 - verify live
Switch to change the interval between ping requests (Linux)	-i
Switch to restrict to IPv4 (Linux)	-4
Switch for more verbose output (Linux)	-v

Practical commands

ping bbc.co.uk
ping muirlandoracle.co.uk
man ping

The DNS address and ping result can vary over time. A host may also block ICMP, so no response does not always mean the host is offline.

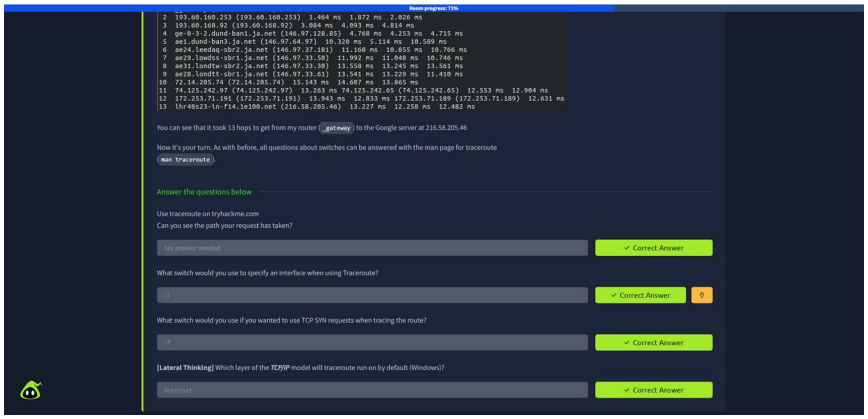
```
C:\Users\Kunal>Ping muirlandoracle.co.uk

Pinging muirlandoracle.co.uk [217.160.0.152] with 32 bytes of data:
Reply from 217.160.0.152: bytes=32 time=146ms TTL=44
Reply from 217.160.0.152: bytes=32 time=146ms TTL=44
Reply from 217.160.0.152: bytes=32 time=146ms TTL=44
Reply from 217.160.0.152: bytes=32 time=146ms TTL=44

Ping statistics for 217.160.0.152:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 146ms, Maximum = 146ms, Average = 146ms
```

Task 6 - Networking Tool: Traceroute

traceroute helps show the route packets take toward a destination by eliciting responses from intermediate network hops. Some hops may not respond, so * entries are possible.



Question	Answer
Can you see the path when tracing tryhackme.com?	Run it in the lab; the path and responding hops vary.
Switch to specify an interface	-i
Switch to use TCP SYN requests	-T
TCP/IP layer used by Windows tracert by default, according to the room	Internet

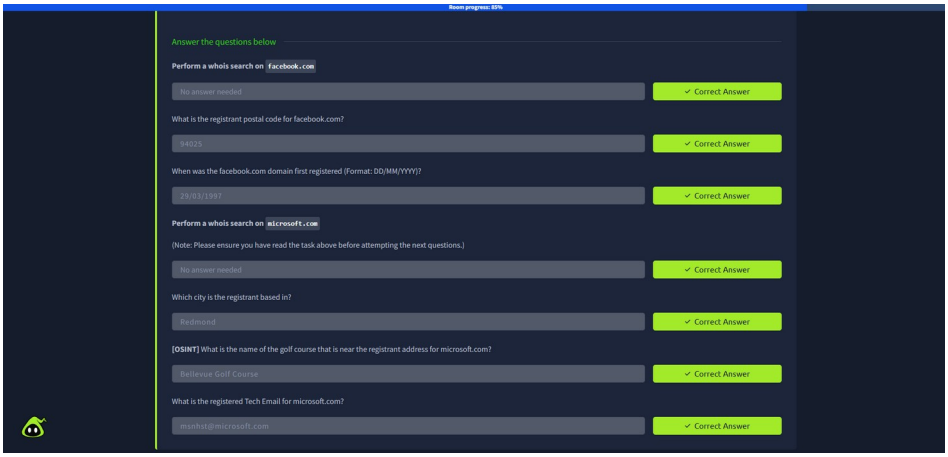
Task 7 - Networking Tool: WHOIS

WHOIS provides registration-related information about domain names, such as registrar data, registration dates, name servers, and registrant details when publicly available.

whois facebook.com

whois microsoft.com

If whois is not installed in the Linux environment, use the web WHOIS tool provided by the room or an approved public lookup service.



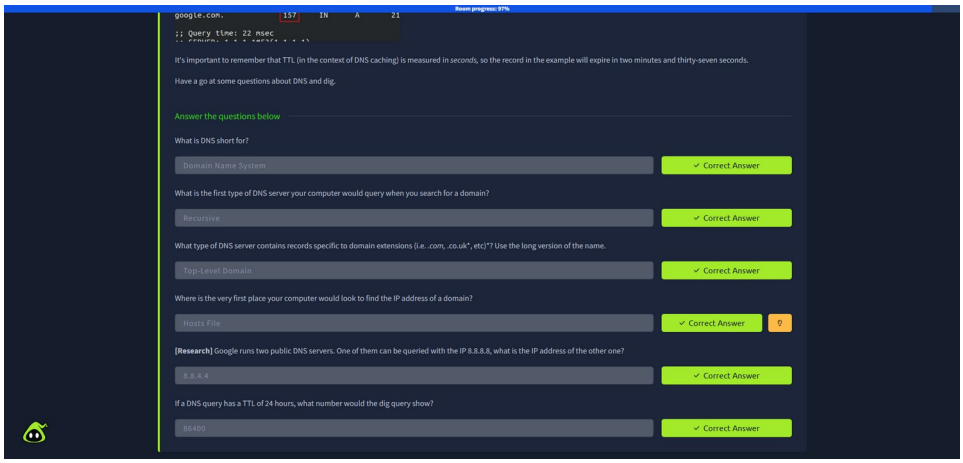
Answers reported for this room

Question	Answer
Registrant postal code for facebook.com	94025
First registration date for facebook.com (DD/MM/YYYY)	29/03/1997
City associated with the microsoft.com registrant	Redmond
Nearby golf course in the room's OSINT exercise	Bellevue Golf Course
Registered technical email for microsoft.com in the room's dataset	msnhst@microsoft.com

WHOIS information can change or be redacted by registrars. Use the output and instructions in your live room if the values differ.

Task 8 - Networking Tool: dig and DNS

DNS (Domain Name System) maps domain names to resource records such as IP addresses. A resolver usually checks applicable local sources and caches before querying other DNS servers.



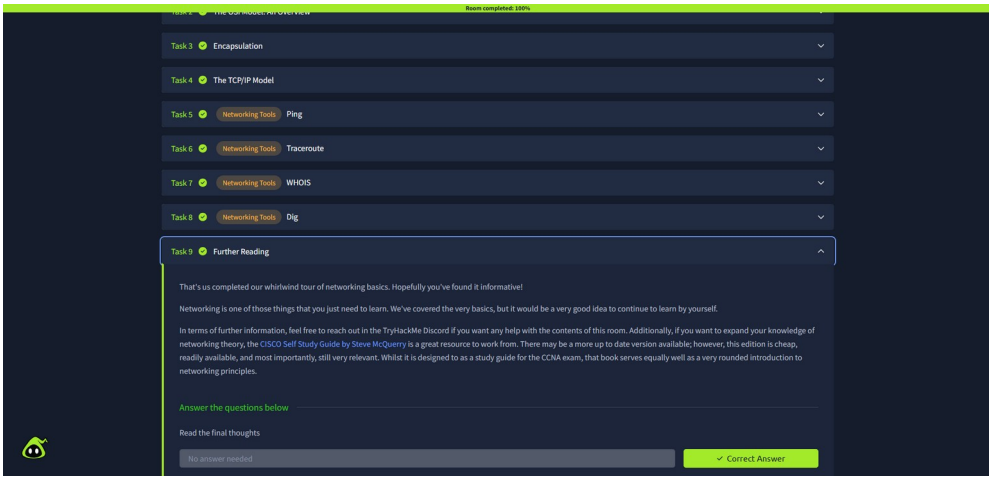
Answers

Question	Answer
What does DNS stand for?	Domain Name System
First type of DNS server queried in the room's example	Recursive
Server type containing records for extensions such as .com	Top-Level Domain
First local place checked in the room's described lookup order	Hosts File
Google's other public DNS address besides 8.8.8.8	8.8.4.4
TTL in seconds for 24 hours	86400

Note on lookup order: The room's explanation describes checking the hosts file before the DNS cache. Some walkthroughs phrase this answer as “local cache”; use the wording that matches the live room's current expected answer.

Task 9 - Further Reading

Read the room's final thoughts and mark the task complete in TryHackMe. No separate technical answer is required; follow the room's completion instructions.



3. Key Learning Outcomes

- The OSI model provides seven conceptual layers; TCP/IP commonly uses four layers.
- TCP is connection-based and uses the SYN, SYN/ACK, ACK handshake.
- Encapsulation adds protocol information; de-encapsulation removes it at the receiver.
- ping helps test reachability; traceroute helps inspect a network path.
- whois reveals available domain registration information.
- DNS resolves domain names, while dig can query DNS records and show TTL values.
- Tool output depends on environment, network policy, and current DNS/registration data.